

Chapter 45 — Posterior Urethral Valves

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Posterior urethral valves (PUV) are the commonest cause of bladder outlet obstruction in infant males. The condition is readily diagnosed and treated by a straightforward initial surgical intervention, yet confers profound long-term consequences. **Obstruction is suspected antenatally in most, and pathognomonic postnatal imaging confirms it.** Endoscopic ablation or urinary diversion relieves the obstruction, but **the embryological insult to bladder and kidney is not relieved**, and these infants require extended joint care from urologists and nephrologists.

Description

- **Morgagni recognised a valve-like posterior urethral obstruction in 1769**, confirmed by **Langenbeck in 1802** — both from post-mortem dissection.
- **Hugh Hampton Young gave the first endoscopic diagnosis and coined the term "posterior urethral valves" (1919).**
- **Randall performed the first endoscopic resection of valves in 1920** (published 1921). His description still matches the modern endoscopic view: **markedly dilated prostatic urethra, raised vesical neck, relaxed bladder orifice, deep pittings in either lateral prostatic urethral wall, and a fine frenulum extending distally from the verumontanum for about 1 cm**, dividing into a valve on either side that rises from floor to side wall.

YOUNG CLASSIFICATION — THE THREE TYPES COMPARED

	Type 1	Type 2	Type 3
Share	95% of PUV	Not definitively reported since the earliest descriptions	5–10% of outlet obstruction
Anatomy	Leaflets arise from the verumontanum , course anteriorly , fuse in the midline just proximal to the external striated sphincter	From verumontanum, extending posteriorly and superiorly to the bladder neck	Annular ring with a small central opening, attached to the entire urethral circumference; resembles congenital urethral stricture
Obstructing	Yes	No	Yes, but for the central aperture
Origin	Hypertrophied inferior urethral crest , from insertion of the distal Wolffian duct ends into the anterolateral cloacal walls	Probably not a real entity	Persistent urogenital membrane after the uorectal septum divides the cloacal membrane
Status	The lesion treated in practice	Stephens found none on cystoscopy in a large series (1955) and reinterpreted the appearance as the secondary bladder-neck effect of more distal obstruction	Contentious but accepted

- **What it is for:** separating the one lesion that is treated from two that are contested. **The trap is treating type 2 as a real obstructing entity.**

EMBRYOLOGICAL THEORIES OF POSTERIOR URETHRAL VALVES

Theory	Mechanism	Standing
Mucosal fold hypertrophy	Hypertrophy of normal urethral mucosal folds obstructs the lumen	Earliest theory , superseded

Theory	Mechanism	Standing
Cloacal remnant	Cloacal remnants produce valve-like leaflets after division by the urogenital membrane	Extension of the above
Wolffian/Müllerian maldevelopment	Lowsley (1914, autopsy) — abnormal development of the Wolffian and Müllerian ducts	Historical
COPUM — congenital obstructing posterior urethral membrane	A persistent, oblique urogenital membrane , perforated by the catheter or feeding tube passed in the immediate postpartum period, converting a membrane into apparent leaflets	Provocative but unproven ; corroborative evidence lacking

- **No consensus embryological account exists.** The pattern across the rows is a steady retreat from "the valve is a structure" toward "the valve is an artefact of how the membrane is first crossed".

Concept — *the classification is historical rather than biological: only type 1 is both common and obstructing, type 3 is a circumferential membrane rather than a valve, and type 2 survives in the nomenclature as a misinterpreted secondary change.*

Genetic Basis of Posterior Urethral Valves

- **The majority of cases are sporadic**, but sibling occurrence means inheritance may be **autosomal recessive, X-linked recessive, or multifactorial**.
- **Chromosome 11 is the recurring locus** — affected siblings shared a partial duplication of the **short arm of chromosome 11**; chromosome 11 with chromosome 1 implicated elsewhere.
- **Copy number variants (CNVs) identify candidate gene regions in the majority of isolated PUV cases examined**, and CNV burden carries phenotypic weight — **a higher nadir creatinine and greater likelihood of progression to stage V chronic kidney disease (CKD)**.
- **Whole-genome sequencing maps PUV loci to TBX5 (transcription factor) and PTK7 (planar cell polarity gene)** — both encoding proteins detectable in the developing human urinary tract.

Concept — *the genetics remain descriptive rather than diagnostic, but the copy number variant work matters clinically because it links genotype to renal prognosis rather than to the presence of the valve.*

EPIDEMIOLOGY

- **Congenital urinary tract anomalies affect up to 1 in 500 pregnancies**, and **obstructive uropathy accounts for most of them**.
- **Lower urinary tract obstruction (LUTO) has an incidence of 2.2 per 10,000 births**, of which PUV is the **commonest pathology at 1.4 per 10,000**, followed by **urethral atresia**, then **prune-belly syndrome**.
- **Prevalence ratio of PUV is 2.10 per 10,000 births** — defined as **affected births divided by total live births plus stillbirths**, a denominator chosen because birth incidence depends on the unreliable counting of early spontaneous pregnancy loss.
- **Prevalence is significantly higher in Black and minority ethnic groups than in white Europeans**.

Concept — *incidence and prevalence are not interchangeable here: the prevalence ratio exists precisely because antenatal demise makes true incidence uncountable, and quoting either figure without its denominator is the error.*

PATHOPHYSIOLOGY

Bladder

- **Detrusor contractility is reduced in PUV bladders**, on both electrical field stimulation and carbachol stimulation of bladder strips, attributed to **replacement of detrusor smooth muscle by fibrosis**.
- **The obstruction produces the triad seen on voiding cystourethrography (VCUG): dilatation of the posterior urethra, hypertrophy of the bladder neck, and flattening of the verumontanum with dilated ejaculatory ducts — pathognomonic**, and these changes largely regress after valve ablation.

Fig. 45.2. Contractile responses and histology of normal and PUV bladder tissue. (A) Frequency–response curves for electrical field stimulation. (B) Concentration–response curves to carbachol. (C) **Elastin van Gieson stain** of normal and PUV bladder sections — **orange = smooth muscle, red = collagen**; scale bar 200 µm. From Johal N, Cao K, Arthurs C, et al., *Journal of Pediatric Urology*, 2021.

Valve Bladder Syndrome — the vicious cycle

- **Mitchell coined the term in 1982**, observing that bladder filling and emptying were intricately related to the degree of renal pelviectasis and to overall renal function.
- The cycle, as a sequence: **fetal obstruction → bladder hypertrophy → high voiding pressures maintain emptying (compensated phase) → progressive wall remodelling and fibrosis with less efficient contraction → falling voiding pressure and rising post-void residual (PVR) → urine volume rises (normal growth, latent polyuria of dysplasia, raised glomerular pressure from poor emptying) → urine stored in the upper tracts → that bolus refills the partially emptied bladder → the bladder is never empty for any sustained period → decompensation → greater PVR, overflow incontinence, further renal damage.**
- **Incontinence in a boy with PUV after a period of normal voiding is a red flag, not a continence problem** — One series found a higher incidence of renal deterioration in boys presenting with incontinence after age 5 **even when infant renal function had been normal**.

Concept — *the valve bladder is a compensation failure, not a persistent obstruction: the same high-pressure hypertrophy that protects emptying in infancy is what fibroses the detrusor and drives decompensation later, so relief of the outlet does not close the cycle.*

Upper Urinary Tract

- **Sustained elevation of intravesical storage pressure over prolonged intervals is transmitted to the ureter, renal pelvis and ultimately the glomerulus**, producing architectural and functional change at every ascending level.
- **Severe hydroureteronephrosis is the clinical marker of that transmission**; increased echogenicity, parenchymal thinning with cortical cysts, and loss of corticomedullary differentiation imply significant renal dysplasia.
- **Ureteric dilatation has three contributors: direct pressure transmission from the dysfunctional bladder, vesicoureteric reflux (VUR) — present in up to 70% of PUV patients — and polyuria from progressive renal damage and congenital dysplasia.**
- **Chronic ureteric dilatation causes wall thickening, loss of peristalsis and loss of mucosal coaptation**, raising the risk of urinary stasis, infection and further pressure rise in the renal units.

Renal Dysfunction

- **Two separate but interrelated aetiologies: obstructive uropathy from the valves, and pre-existing renal dysplasia.**

Evidence for the obstructive component

- **Fetal sheep ureteric obstruction models:** hydronephrosis develops rapidly, but **irreversible dysplastic architectural change is present by term**.
- **Apoptosis and increased oxidative stress** in murine obstructed kidneys may mediate dysplasia in the face of outlet obstruction.
- **Raised pressure damages renal tubular luminal cells → poorly concentrated urine;** obstruction additionally **reduces medullary blood flow → loss of the medullary concentration gradient → significant polyuria and the post-obstructive diuresis that follows catheter placement** in infants with PUV.

Evidence for the dysplastic and renin-angiotensin component

- **Severe dysplasia correlates with decreased renin-angiotensin system activity** and with angiotensin II type 1 receptor polymorphisms.
- **Plasma renin activity rises before the conventional markers of renal damage** — before creatinine rise, renal scarring or reduced glomerular filtration rate (GFR).
- **Angiotensin II damages the kidney twice over:** haemodynamically, by altering glomerular flow, and **profibrogenically, by inducing transforming growth factor- β 1 (TGF- β 1) and tumour necrosis factor- α (TNF- α).**
- **TGF- β 1 and TNF- α are candidate biomarkers:** they **fall as renal function improves after valve ablation** and rise with worsening outcome.
- **Angiotensin-converting enzyme inhibitors (ACEIs) reduce fibrosis experimentally and reduce TGF- β 1 and TNF- α expression clinically** — a potential role in selected children with PUV.
- **ACE2 gene expression is significantly higher, and angiotensin II type 1 receptor gene expression lower, in PUV than in other renal dysplasias and in controls** — supporting a congenital dysplastic, not purely obstructive, mechanism.
- **Countervailing finding:** fetal sheep with **partial** bladder outlet obstruction show **increased renin and angiotensin messenger RNA with upregulated AT2 receptor expression** — the opposite direction.

Concept — *the governing asymmetry is dysplasia versus obstruction: obstruction is relievable and its renal injury partly reversible, dysplasia is fixed at birth and sets the ceiling on long-term function, and the contradictory renin-angiotensin data exist because the two processes are being measured together in the same kidney.*

Vesicoureteral Reflux and Dysplasia

- **Hoover and Duckett** observed that **high-grade VUR in PUV is usually ipsilateral to a poorly functioning renal unit, with preserved function contralaterally**, and hypothesised a **"pop-off" mechanism** — the dysplastic refluxing kidney acting as a **pressure reservoir** protecting the opposite kidney. They coined **vesicoureteral reflux and dysplasia (VURD)**.
- **Longer follow-up has refuted the protective claim.** In longer-term follow-up, most VURD children with a normal serum creatinine in the second year of life had lost that normal value by 8 to 10 years, and **GFR was already abnormal in the majority within the first 2 years** — creatinine had simply not yet declared it.
- Children with valve bladder and a pop-off mechanism were **more likely to have unsafe initial urodynamics, higher voiding pressures, voiding dysfunction and a requirement for clean intermittent catheterisation (CIC)** than those without.
- **The contralateral, non-refluxing unit itself carries a high risk of congenital scarring.**

- **VURD must never generate a false sense of security** — follow-up is as close as for any other PUV cohort.

Concept — *VURD is the classic false reassurance: the observation is real, the inference of contralateral protection is wrong, and the reason is that a normal creatinine in a growing infant conceals an already abnormal glomerular filtration rate.*

TABLE 45.1 DAMAGE CAUSED BY POSTERIOR URETHRAL VALVES

Organ	Lesion	Effect	Natural history
Lung	Pulmonary hypoplasia	Respiratory failure	May be fatal in the newborn; limited long-term data in survivors
Kidney	Obstructive uropathy	Reversible renal insufficiency	Usually improves with initial treatment, but can recur with bladder dysfunction
Kidney	Dysplasia (glomerular injury)	Irreversible renal insufficiency	Permanent damage limiting growth; progressive renal failure and hypertension
Kidney	Tubular injury	Inability to limit sodium and water loss	Progressive with age; nephrogenic diabetes insipidus
Bladder	Poor sensation, hypercontractility, low compliance, eventual myogenic failure	Incontinence and poor emptying	Lifelong; changes with age and management
Ureters	Poor contractility, inability to coapt and transport urine	Stasis, infection	Many improve initially, but most retain chronic hydronephrosis

- **What it is for:** it is the **reversibility map** of the disease, organised by **which lesion ablation can undo and which it cannot**.
- **The pattern:** obstructive uropathy is reversible; dysplasia and tubular injury are not; the bladder alone is described as lifelong and *changing*.
- **The trap:** reading "hydronephrosis persists" as treatment failure — for the ureter, persistence is the expected natural history, not residual obstruction.

KEY POINTS

- The renal dysfunction, vesicoureteric reflux and voiding dysfunction seen in children with PUV are **mediated by a dysfunctional bladder**.
- Renal impairment in PUV is caused by **coexisting renal dysplasia and obstructive uropathy**.
- **VURD syndrome confers no protective benefit** on long-term renal prognosis.

DIAGNOSIS

Antenatal Diagnosis

- **Antenatal detection continues to improve** with better fetal ultrasound and magnetic resonance imaging (MRI).
- **Detection prepares the team and the parents, but there is scant evidence that it improves outcomes for the child.**
- **A meta-analysis found no demonstrable improvement in progression to CKD or renal replacement therapy** attributable to antenatal diagnosis, the literature being limited by heterogeneous cohorts and significant bias.

- The counterintuitive result **underscores the limits of current technology in both diagnosis and prediction of kidney outcome**, and the need for standardised, multimodal antenatal care pathways.

Concept — *antenatal diagnosis changes when the disease is known about, not what the kidney has already become: the dysplastic injury is established in utero, so detection cannot be expected to alter renal destiny.*

Ultrasonography

- PUV is detected in approximately 1 in 1250 ultrasound screenings, accounts for 10% of significant antenatally detected genitourinary disease, and afflicts one third of surviving infants with bilateral renal disease.
- Corroborating signs: oligohydramnios, and the dilated posterior urethra producing the "keyhole sign".
- Increased renal echogenicity is a reliable indicator of renal damage, not merely of obstruction.
- The mandate is nonetheless fixed: a thickened, enlarged bladder with bilateral ureterectasis and pelvicaliectasis, with or without oligohydramnios, in a male fetus requires early postnatal ultrasound and VCUG before discharge from the postpartum unit.
- About half of PUV births are diagnosed antenatally; of the remainder, half present in the neonatal period and half with urinary tract infection (UTI).

Concept — *antenatal ultrasound reliably detects lower urinary tract obstruction but poorly identifies its cause, which is why the antenatal finding triggers a postnatal protocol rather than an antenatal diagnosis.*

Fetal Magnetic Resonance Imaging

- An adjunct, increasingly available at major centres, not a replacement.
- Assesses the same three parameters as ultrasound: degree of urethral dilatation, bladder distension with wall thickening, and reduced amniotic fluid.
- In early gestation it altered the initial sonographic diagnosis in a substantial minority of cases.

Postnatal Diagnosis

Ultrasonography

- In the newborn, free fluid around one or both kidneys implies a urinoma.
- Bladder wall is regarded as thickened at more than 3 mm when distended, or more than 5 mm when empty.

Voiding Cystourethrography

- Timing sequence: renal and bladder sonography first → VCUG as soon as the infant is haemodynamically stabilised and able to undergo a contrast study → before discharge.
- High-grade VUR is present in approximately half of patients at diagnosis.
- The voiding-phase appearance is the pathognomonic sequence: contrast crosses a hypertrophied, elevated bladder neck → enters a grossly dilated posterior urethra → funnels abruptly at a transverse membrane or cusp, representing the obstructing leaflets seen at cystoscopy.
- Technique, as a sequence: insert a 5- to 8-Fr feeding tube → if it curls in the capacious posterior urethra or at the hypertrophied bladder neck, use a coudé catheter to advance into the bladder → during the voiding phase, withdraw the catheter gradually to a point distal to the posterior urethra so that segment is seen unobstructed. The same withdrawal rule applies where a feeding tube is already in situ.

Concept — *the whole diagnostic yield of the voiding cystourethrogram sits in the voiding phase at the point of abrupt calibre change, which is why a catheter left across the posterior urethra produces a technically complete but diagnostically worthless study.*

Voiding Urosonography

- **Second-generation ultrasound contrast permits dynamic, radiation-free imaging of the lower urinary tract.**
- **Evidence now supports ceVUS for urethral imaging** — reliable diagnosis of PUV and of other urethral pathology such as **urethral diverticulum**.
- **Increasingly used as part of a concerted effort to reduce radiation exposure in children, and useful when performed by experienced radiologists.**

Radionuclide Renal Scan

- **Its purpose is quantification of differential renal function, not diagnosis of the valve.**
- **Cortical deficits on the scan may imply renal dysplasia.**
- **Timing is dictated by tubular maturity:** the study is performed at about **2 to 3 months of age**, so the tubules can process the radionuclide accurately.
- **Mercaptoacetyltriglycine (MAG3) is the agent of choice** for assessing functional contribution.
- **The trap is interpretive: delayed clearance of tracer from a dilated collecting system is readily misread as ureterovesical junction obstruction.**

Laboratory Evaluation

- **Neonatal values reflect maternal biochemistry** and must be interpreted with caution.
- **After 48 hours the placentally transmitted maternal contribution clears** and the infant's own baseline renal function becomes apparent; subsequent values and **trends** are what infer effective renal function.
- **Nadir creatinine at 1 month, 6 months and 1 year is the key prognostic instrument**, and may be plotted to assess the immediate response to treatment in the neonatal period.
- **The caveat is that the serum creatinine plateau may not be reached even in unaffected children until well into the first year** — between roughly day 65 and day 220 of life.

Concept — *every biochemical number in the newborn with valves is a moving target: the first 48 hours belong to the mother, the plateau belongs to the second half of the first year, and only the nadir creatinine trend carries prognostic weight.*

IMAGING MODALITIES COMPARED

Modality	Detects	Strength	Limitation
Antenatal ultrasound	Thick dilated bladder, bilateral hydroureter and pelvicaliectasis, keyhole sign, oligohydramnios, raised renal echogenicity	Sensitivity 95% / specificity 80% for LUTO; universally available	Accuracy for the specific cause falls to 50% against atresia, prune-belly, high-grade VUR, megaureter
Fetal MRI	Same parameters plus pulmonary hypoplasia and renal cystic change	Alters the sonographic diagnosis in a substantial minority in early gestation	Still cannot establish the cause of LUTO ; cystic change does not track dysplasia
Postnatal ultrasound	Thick-walled bladder, diverticula, dilated posterior urethra, high bladder neck, keyhole sign, urinoma	First postnatal study; defines upper tract and dysplasia	Not diagnostic of valves
VCUG	Trabeculated bladder with diverticula, VUR, elevated bladder neck, dilated posterior urethra, abrupt funnelling at the valve	Definitive diagnostic study	Radiation; requires haemodynamic stability; technique-dependent
ceVUS	Reflux and urethral pathology including PUV and diverticulum		Not yet the standard of care

Modality	Detects	Strength	Limitation
		No ionising radiation; sensitivity reportedly exceeds VCUG for reflux	

- **What it is for:** staging the pathway by what each test can exclude rather than confirm. **Ultrasound establishes that there is obstruction; only the voiding study establishes what is obstructing.**

KEY POINTS

- **Antenatal diagnosis is primarily by fetal ultrasound**, sometimes aided by fetal MRI; a thickened dilated bladder with upper tract dilatation and oligohydramnios has high sensitivity for PUV.
- **Postnatal diagnosis is by ultrasound, confirmed by fluoroscopic VCUG**; ceVUS is emerging as a good alternative.
- Biochemical evaluation centres on electrolytes and creatinine; **nadir creatinine is an effective prognostic indicator** for long-term kidney function.

CLINICAL PRESENTATION AND INITIAL MANAGEMENT

Antenatal Presentation and Management

- A significant proportion of children with PUV now reach the urologist antenatally, for prenatal advice and management.
- The three considerations that govern antenatal decision-making are the presence of oligohydramnios, the degree and laterality of hydronephrosis, and the gestational age.

Selection for antenatal intervention

- Antenatal intervention was popularised in the mid-1980s, enabled by advances in fibreoptics and endoscopic miniaturisation.
- Ultrasound remains the mainstay of antenatal diagnosis, but fetoscopy offers better sensitivity — at the cost of the considerable risks of intervention.
- **Historical criteria for offering intervention:** oligohydramnios, a dilated bladder, and severe hydroureteronephrosis, **without renal cortical cystic lesions**, in a fetus with a **normal karyotype**.
- The Society of Fetal Urology convened a multidisciplinary panel in 2016, proposing a **standardised severity classification of LUTO**.
- A published stratification names the four features predicting the most favourable outcome from intervention: oligohydramnios but not anhydramnios; severe hydronephrosis; favourable fetal urine biochemistry; absence of renal cortical cysts.
- The counterargument is for **selective, cautious intervention**: some children do well despite significant obstruction, and others are carried through the intrauterine period by intervention only to fare badly afterwards.

FETAL URINE BIOCHEMISTRY — FAVOURABLE PROGNOSTIC THRESHOLDS

Analyte	Favourable value (sample after 20 weeks)
Sodium	< 100 mEq/L
Chloride	< 90 mEq/L
Osmolality	< 200 mOsm/L

Analyte	Favourable value (sample after 20 weeks)
β 2-microglobulin	< 6 mg/L

- **A single sample is inferior to serial sampling: triple serial sampling at 48- to 72-hour intervals** is the more accurate proxy for irreversible renal damage, with **sequential rises across samples portending an unfavourable prognosis**.
- **Fetal urine biochemistry before 23 weeks significantly predicts renal outcome** — β 2-microglobulin the **most sensitive marker**; sodium and calcium the **most specific** but less sensitive.
- **Contrary evidence:** a systematic review found that fetal urine indices did not accurately predict poor renal function. The results were not universally replicated.

Concept — *the fetal urine argument rests on the dysplastic tubule's inability to reabsorb: low solute means a kidney still working, and it is the trajectory across serial samples, not any single threshold, that carries the prognosis.*

Vesicoamniotic shunting

- **Shunting treats oligohydramnios, not the kidney** — the potential benefit is **ameliorative for pulmonary function**, and it **represents the first stage of fetal intervention**; several hundred procedures are reported.
- **Fetal sheep models show that restoration of amniotic fluid volume prevents lung hypoplasia**, though absence of controlled studies constrains conclusions.
- **A systematic review showed a survival advantage after shunting, but without randomised trials.**

PLUTO — Percutaneous vesicoamniotic shunting versus conservative management for Lower Urinary Tract Obstruction (2013)

- **Design:** randomised comparison of vesicoamniotic shunting against conservative management in fetal LUTO.
- **Limited by poor recruitment and pregnancy terminations** — only 12 live births in each arm.
- **Result:** a trend toward improved survival at 28 days in the shunted group, but overall survival was very poor in both arms — only two infants survived to 2 years of age with normal renal function.
- **High mortality from pulmonary hypoplasia**, and **greater pregnancy loss in the shunt arm** from procedure-related complications and early rupture of membranes.

SHUNT DEVICES COMPARED

	Harrison shunt (traditional)	Somatex intrauterine shunt
Design	Double-pigtail shunt	Nitinol wire mesh stent
Calibre	5 Fr, internal diameter 0.97 mm	Internal diameter 2.6 mm
Performance	Reference device	Less likely to dislocate; better renal outcomes in comparative study

- **What it is for:** the newer device wins on calibre and fixation, not on any change in the underlying biology.

Fetal cystoscopy and fetal surgery

- **No PLUTO-equivalent randomised evidence exists for the more complex interventions.**
- **Technique, as a sequence:** 1.0-mm fetoscope → antegrade transuterine percutaneous incision into the fetal bladder → entry into the posterior urethra through the bladder neck → perforation of the obstructing membrane with an Nd:YAG laser or cauterising wire.
- **A series of 14 fetal procedures** — antenatal valve ablation, vesicoamniotic shunting, cutaneous ureterostomy and vesicostomy: **fetal mortality 43%**, six infants dying of premature delivery and respiratory failure, and **most surviving children in renal failure at a mean of 11.6 years**.

- **That fetal loss rate is a mandatory element of prenatal counselling** before any intervention offering potential, unproven, near-term survival benefit in a select group of pregnancies.

FETAL INTERVENTION COMPARED

	Vesicoamniotic shunting	Fetal cystoscopy ± ablation
Target	Amniotic fluid volume → lung development	The obstructing membrane itself
Access	Percutaneous shunt, bladder to amniotic cavity	1.0-mm fetoscope, antegrade transuterine, Nd:YAG laser or cautery wire
Randomised evidence	PLUTO — trend to 28-day survival only	None
Comparative outcome	No difference between the two in fetal survival, kidney function or urological outcome	
Pregnancy loss	40–50%	40–50%
Renal benefit	Not demonstrated	Not demonstrated

- **What it is for:** both procedures converge on the same verdict, so the choice between them is technical rather than prognostic.

Concept — *fetal intervention can plausibly rescue the lung and cannot rescue the kidney: amniotic fluid restoration addresses a mechanical deficiency still in progress, while renal dysplasia is already established — which is why every comparative study converges on no renal benefit at a 40–50% pregnancy loss.*

Parental counselling

- **A defined role for the paediatric urologist:** educate parents on the **putative diagnosis, possible outcomes, alternative diagnoses and the expected course of the remainder of the pregnancy.**
- **Direct delivery to an institution equipped for immediate neonatal surgical management.**
- **Frame the prenatal visit as the start of a lifelong relationship** — continuity of appropriate expertise from infancy into adulthood.

KEY POINTS

- **Up to 50% of infants with PUV are diagnosed antenatally.**
- **Severity of obstruction is objectively assessed by amniotic fluid volume, degree of renal dysplasia, and fetal urinary markers.**
- **Vesicoamniotic shunting may confer a survival benefit in selected cases, but no clear benefit in the risk of renal failure.**

Postnatal Management

- **Serious comorbidities may dominate the presentation:** pulmonary hypoplasia and the **physical stigmata of oligohydramnios** — **Potter facies, club feet, deformed hands, poor visceral muscle tone** — and may demand intensive initial management.
- **Clinically the infant may void with difficulty, with a weak or intermittent stream.**
- **Technique for drainage, as a sequence:** pass a 5- or 7-Fr feeding tube (or similar-calibre catheter) per urethra → if it curls in the dilated posterior urethra or is impeded by the hypertrophied, elevated bladder neck, use a coudé-tip catheter, or curl the feeding-tube tip dorsally over a stylet → apply posterior digital rectal pressure to guide the tube anteriorly through the dilated prostatic urethra and high bladder neck → confirm intravesical placement by bladder sonography or, in some cases, a one-shot cystogram.

- **Minimise balloon filling:** in a small, hypertrophic bladder a filled balloon induces bladder spasm and may occlude the ureteric orifices.
- **Hyponatraemia may be present, sometimes triggered by UTI;** it is readily corrected with sodium supplementation and is attributed to transient pseudohypoaldosteronism.

Concept — *the first catheter is the whole of initial management, and every technical detail exists because the anatomy that obstructs the valve also obstructs the catheter: the dilated posterior urethra that curls the tube and the elevated bladder neck that deflects it.*

Pulmonary Hypoplasia

- Despite the urological focus on bladder and kidney, pulmonary hypoplasia is the most profound complication and the commonest cause of perinatal mortality in severe LUTO.
- The cyanotic neonate requires complex ventilatory support, and this is typically the factor delaying definitive surgical treatment of the valves.
- Aetiology is unclear and likely multifactorial, the proposed mechanism being a sequence: oligohydramnios plus grossly hydronephrotic kidneys and a dilated bladder → increased uterine pressure on the fetus → reduced diaphragmatic expansion → reduced lung volume and growth → impaired development of the fetal pulmonary tree, which requires intraluminal pressure, volume and flow, together with cellular signalling to the developing alveoli.

Urinomas

- A urinoma complicates 3% to nearly 10% of PUV cases; detection rose to around 15% after the introduction of renal ultrasonography in 1979.
- Two distinct extravasation patterns: forniceal rupture → fluid trapped within the renal capsule, appearing on ultrasound as distorted renal parenchyma; and transperitoneal transudation of fluid, or bladder rupture → neonatal ascites.
- VCUG or radionuclide cystography may delineate the site of leakage in ascites, but the cause is often not determinable.
- Management is almost always indirect: restore urinary flow and treat the proximate cause of LUTO, and the fluid reabsorbs.
- Percutaneous drainage or tapping is reserved for ascites causing respiratory distress, severe abdominal distension, or other clinical symptoms.

THE URINOMA POP-OFF DEBATE — THE TWO POSITIONS

Urinoma is protective	Urinoma is harmful or neutral
Acts as a pop-off mechanism , reducing renal dysplasia on that side	Especially when retained within the renal capsule, it compresses the kidney and impairs ipsilateral function
Some series show globally preserved renal function , including a long-term index of renal severity	A harbinger of worse renal prognosis , or no bearing at all on long-term function

- **What it is for:** the debate does not change management, because relief of the obstruction is the treatment on either reading.

Concept — *the urinoma reprises the VURD argument in miniature: an apparently decompressive lesion is claimed to spare the kidney, the evidence is contradictory, and management does not turn on the debate.*

Delayed Presentation

- Delayed presentation is assumed to be uncommon in the era of antenatal ultrasound, and is not.

- The majority of children undergoing valve ablation (62%) presented with something other than prenatal hydronephrosis or oligohydramnios; up to 64% of these had a normal prenatal ultrasound, and about a tenth presented with UTI, voiding complaints and acute renal failure.
- 43% presented after 6 months of life.
- Features that should raise suspicion in a male: lower urinary tract symptoms, especially recurrent UTI; also overflow incontinence, gross haematuria, renal dysfunction, and, less commonly, ejaculatory dysfunction.
- The screening step is renal ultrasound — the telltale findings being bladder wall thickening and distal ureteric dilatation — with VCUG for confirmation.

Concept — a normal prenatal ultrasound does not exclude valves in a boy, which is why the diagnosis remains a clinical suspicion driven by symptoms rather than an antenatal one.

KEY POINTS

- An antenatal finding of a thickened bladder and bilateral ureterectasis should be evaluated with early postnatal ultrasound and VCUG.
- VCUG in PUV shows bladder wall irregularity, a hypertrophied and elevated bladder neck, and a dilated, elongated posterior urethra.
- The commonest cause of early mortality in infants with PUV is pulmonary hypoplasia.

SURGICAL INTERVENTION

Valve Ablation

- Cystoscopy with valve ablation is the preferred initial surgical option in any neonate diagnosed with PUV.
- The treatment goal is to restore urethral urine flow and enable normal cyclic filling and emptying of the bladder — superior to urinary diversion with passive drainage.
- Experimental support for cycling: a diversion-and-undiversion model showed the changes occurring in a bladder prevented from cycling, and a fetal sheep model developed increased expression of extracellular matrix elements and apoptosis after high diversion.
- Where the infant is small, premature or of low birth weight, urethral calibre may preclude cystoscopy — primary vesicostomy is then an option; alternatively, sequential urethral dilatation to accommodate a cystoscope over a mean of about a fortnight has been described.

VALVE ABLATION — APPROACHES AND INSTRUMENTS

Approach	Instrument	Notes
Blind / tactile hook	Crochet hook passed retrograde, feeling it catch the obstructing tissue	Innes Williams first described engaging valves with a hook
Insulated hook	Whitaker and Sherwood modification — wire insulated except at the distal hook, 6–7 Fr	Bedside use without general anaesthesia, with a small amount of diathermy
Endoscopic ablation (preferred)	7.5-Fr or 9-Fr infant cystoscope with offset lens, passing a holmium:YAG laser fibre or Bugbee electrode	Incise at the ventral 5 and 7 o'clock positions, with or without a dorsal 12 o'clock incision; a 12 o'clock incision alone may suffice
Improvised ablaters	Wire bent at the tip passed through a 3-Fr ureteric catheter; visually guided Fogarty embolectomy catheter	Alternatives where dedicated devices are unavailable

Approach	Instrument	Notes
Resection	9.5-Fr resectoscope with Collins knife working element	Only in an infant with a normal-calibre urethra; excellent visualisation, stable platform

- The pattern across the rows is a trade-off between visualisation and urethral calibre: the 7.5-Fr cystoscope gives limited visualisation once the ablating device occupies the working channel, while the infant resectoscope gives excellent visualisation but may be too large, risking meatal injury when the meatus is small.
- Where the resectoscope is desired but the urethra too small, the PADUA approach — progressive augmentation by dilation urethra anterior — serially up-sizes a urethral drainage tube from 5 Fr to 8 or 10 Fr over 3 to 5 days.
- Valves are thin and minimally vascular: aggressive resection is to be avoided. Hot-loop resectoscope ablation, used mainly in older children, carries a higher risk of urethral stricture.

Postoperative course

- A urethral catheter is typically left for at least 24 hours and need not be kept longer, though longer catheterisation is common while renal or respiratory parameters are monitored in intensive care.
- A VCUG must be repeated within 1 month of ablation to confirm that the valves are no longer visible.
- Markers of adequate ablation: improvement in bladder wall thickness, reduced upper tract dilatation, and demonstrably reduced posterior urethral dilatation. Immediate signs of diminished bladder pressure, with some improvement in renal dilatation and in the volume of reflux, are common.

Concept — the operation is an incision, not a resection: the valve is an avascular leaflet, so the endpoint is relief of the funnel on the follow-up voiding study, and the two commonest errors — over-resection and hot-loop use — both trade a solved obstruction for a urethral stricture.

Bladder neck procedures

- The rationale was anatomical: bladder neck hypertrophy with elevation of the bladder neck proximal to the posterior urethra, and the incomplete emptying persisting on imaging after ablation in some males, prompted interest in transurethral incision of the bladder neck during or after primary ablation.
- Earliest reports were Y-V plasty and bladder neck incision; after Mitchell's elaboration of the valve bladder, bladder neck obstruction was widely accepted as a plausible explanation for persistent dysfunction.
- Bladder neck incision is still offered by some, but adoption is limited by concerns of retrograde ejaculation and by lack of improvement against controls even in short-term pilot studies; the benefit demonstrated in neurogenic bladder has not transferred, and there is a dearth of studies confirming extended durability.
- Botulinum toxin injection into the hypertrophied bladder neck failed to improve voiding dynamics.

Vesicostomy

- It reduces bladder storage pressures and may optimise GFR.
- The old objection — that vesicostomy defunctionalises the bladder and causes long-term loss of compliance — has been refuted: a properly created vesicostomy allows bladder filling and preserves contractile function, because urine must still be expelled through the stoma, albeit at a reduced leak point pressure.
- It is a temporary diversion: it does not alter ultimate clinical outcomes compared with primary ablation, and does not prevent the bladder from serving as an adequate reservoir for renal transplantation.

- **Technique, as a sequence:** 2-cm midline transverse incision midway between the pubic symphysis and umbilicus → separate the rectus muscles → expose the bladder with traction sutures → mobilise the peritoneum cephalad off the posterior wall and dome → identify the dome by isolating the urachus and ligate it → excise the urachus with a small portion of the dome → suture the detrusor to the fascia 1 cm below the edge of the cystotomy.
- **The key operative step is to keep the posterior bladder wall taut, achieved by bringing the dome to the skin — this prevents prolapse of the back wall through the incision.**

Upper Tract Diversion

- **The rationale:** direct decompression of the kidney by **cutaneous ureterostomy or pyelostomy** gives **low-pressure drainage**, and was thought to **protect the upper tract from ureterovesical junction obstruction** caused by a tortuous intramural ureter traversing a thickened, valve-affected bladder. **Loop diversion of ureter or renal pelvis still permits bladder filling and cycling**, mitigating the defunctionalisation concern.
- **The refutation:** long-term follow-up **failed to detect any renal-protective benefit**, and diversion **often requires a complicated undiversion as the child matures**.
- **The decisive observation:** renal biopsies taken at the time of high diversion uniformly show renal dysplasia, while the diversion does not protect from renal failure — implying that **dysplasia occurring in early fetal development, not neonatal surgical strategy, is the primary mediator of renal outcome**.
- **Residual indication:** a rare undertaking in an infant with **complete lower tract decompression yet worsening renal function, increasing upper tract dilatation and possibly sepsis**, undertaken after multispecialty consultation with paediatric nephrology and radiology, and with the explicit understanding that improved renal outcome is not supported by evidence.

THE SURGICAL LADDER COMPARED

	Endoscopic valve ablation	Vesicostomy	Upper tract diversion
Position	Preferred initial intervention in any neonate	Second line — ablation impossible, or decompression not achieved by catheter or ablation	Last resort , rare
Indication	Diagnosed PUV with adequate urethral calibre	Very low birth weight with unusable urethra ; persistent renal or upper tract deterioration	Persistent upper tract dilatation and worsening renal parameters after vesicostomy , ± sepsis
Bladder cycling	Preserved — the physiological goal	Preserved (urine expelled through the stoma at reduced leak point pressure)	Loop diversion permits filling and cycling ; extended defunctionalisation still a risk
Renal benefit	Relieves obstruction	May optimise GFR	No long-term renal-protective benefit demonstrated
Reversal	None needed	Simple closure	Complicated undiversion required

- **What it is for:** the ladder is ordered by **how much bladder cycling each rung preserves**, and the ceiling on all of it is dysplasia.

Concept — *the uniform finding of dysplasia on biopsy at the time of high diversion is why no operation above ablation improves the kidney: the surgery can only change the pressure, never the nephron mass laid down in fetal life.*

Circumcision

- **The risk being addressed is UTI progressing rapidly to pyelonephritis and sepsis**, driven by the associated VUR, incomplete emptying and severe upper tract dilatation.

- **Overall UTI risk in PUV is 50–60%, against about 1% in unaffected males.**
- **Circumcision reduces that risk by 83–92%.**
- **Recommendation: strongly consider prophylactic circumcision in any male diagnosed with PUV, and certainly complete it before entertaining ureteral reimplantation** for frequent febrile UTIs despite conservative measures.
- **A randomised multicentre trial** — circumcision plus antibiotic prophylaxis versus antibiotic prophylaxis alone, with febrile UTI as the primary outcome: **febrile UTI in 3% of the circumcision arm against 49% with prophylaxis alone over 2 years. Circumcision is confirmed as a critical adjunct.**

Nephroureterectomy

- **Historical indication: the non-functioning refluxing renal unit of VURD syndrome**, regarded as a source of infection and sepsis, warranting prophylactic excision.
- **Contemporary position: a focus on bladder emptying and circumcision has reduced UTI incidence sufficiently that nephroureterectomy is rarely considered.**
- **Renal preservation is prioritised even for poorly functioning units contributing moderate polyuria** — polyuria is preferable to anuria.
- **Where frequent UTIs localise to the non-functioning unit and mandate nephrectomy, preserve the ureter** for potential later reconstruction such as **ureteral augmentation**.

Concept — *the ureter of a dead kidney is a graft and its urine is a safety valve against anuria, which is why removing the VURD unit costs more than it saves once infection is controlled by emptying and circumcision.*

KEY POINTS

- **Cystoscopy with valve ablation is the preferred initial treatment for PUV.**
- **Vesicostomy does not inhibit bladder cycling** — the bladder continues to contract — but is reserved for selected cases in which valve ablation is not possible.
- **High urinary diversion offers no renal protective benefit** and requires a complex secondary procedure to reverse.
- **Circumcision should be encouraged prophylactically in any male with PUV**, and especially any male with a history of UTI.

MANAGEMENT OF VESICoureTERAL REFLUX

- **VUR is found on VCUG in 50–80% of infants worked up for PUV.**
- **The governing principle is that reflux here is secondary** — a consequence of obstruction and secondarily elevated bladder pressures — which is precisely why it should dissuade ureteral reimplantation in nearly all cases.
- **Ablation or vesicostomy alone resolves reflux in 25–40% of those refluxing beforehand.**
- **Reflux resolves more readily at lower grades and when the associated kidney has adequate function.**
- **The presence of reflux did not correlate with renal outcomes** — reflux alone is not an indication for intervention.
- **The management target is the bladder, not the ureter: address persistent bladder outlet obstruction, reduce intravesical pressure with anticholinergic treatment, and treat the underlying bladder dysfunction; urodynamic evaluation may guide this.**
- **Where recurrent UTI genuinely mandates intervention, bladder management must precede surgery.**

- **Surgical risks are elevated in this population: higher rates of stricture and persistent reflux** when dilated ureters are reimplanted into thick-walled bladders that have not been properly rehabilitated beforehand.
- Any treatment of reflux in a child with profound bowel dysfunction risks further deterioration of an already compromised renal unit.
- Endoscopic correction carries fewer risks of myogenic disruption and renal deterioration than ureteroneocystostomy in valve patients, but its overall success rate is lower than in children without valves.

Concept — *reflux in valves is a pressure gauge, not a lesion: correcting it surgically without first rehabilitating the bladder treats the reading rather than the pressure, which is why reimplantation is rare and, when unavoidable, follows bladder management rather than preceding it.*

BLADDER DYSFUNCTION AND VALVE BLADDER SYNDROME

- The bladder is the lifelong focus of management, having been exposed to obstruction from its earliest development.
- The bladder initiates the cascade: voiding dysfunction, urinary reflux, and worsening of renal dysplasia and obstructive uropathy — with valve bladder syndrome as the ultimate manifestation.
- Three sequential contractility patterns through childhood: detrusor hyperreflexia in infancy and early childhood → decreasing intravesical pressures and improved compliance in childhood → increased bladder capacity with hypocontractility and atony in adolescence and beyond.
- An essential qualification: the patterns overlap in most children and are not arbitrary or destined — they are **guideposts for monitoring, for choosing management, and for assessing the effect of intervention over the long term.**
- Continence is reached late: males with PUV achieved daytime and night-time continence at around 5.5 years against 2.9 years in peers.
- Clinical history grossly underestimates the problem: after endoscopic ablation, **self-reported incontinence averaged about a fifth of patients, while urodynamically demonstrated bladder dysfunction was present in over half.**
- Uroflowmetry and post-void residual measurement should therefore be routine in follow-up of any toilet-trained child with a history of PUV, rather than relying on examination or questionnaires.

Bladder Management

- Follow-up rests on observation, clinical history and urodynamics, with parent and child education as the critical determinant of success for any behavioural prescription.
- Counsel families not to push toilet training aggressively, and to expect a lag behind the normal population.
- Expected continence outcomes: daytime incontinence in 7–35%; nocturnal enuresis in about one in four children after valve ablation.
- Once toilet training is achieved, the behavioural package is: adequate fluid intake, timed voiding, and double voiding; biofeedback therapy and home pelvic floor exercises are also useful.

PHARMACOLOGICAL OPTIONS — TARGETS AND EVIDENCE

Agent	Target and rationale	Evidence and caution
α -adrenergic blockade	Relieves sphincteric hypertonicity and relaxes the bladder neck — addresses the	Significant reduction in residual volumes in children with high PVR; uncontrolled

Agent	Target and rationale	Evidence and caution
	<i>functional bladder neck obstruction hypothesis</i>	
Oxybutynin (antimuscarinic)	Reduces high voiding pressures and increases small bladder capacity — addresses the <i>detrusor wall thickening from collagen deposition</i> hypothesis	Both parameters improved, but no control group , and detrusor hypercontractility with elevated voiding pressures is a normal neonatal finding ; dose 0.1 mg/kg twice daily
Oxybutynin — randomised trial	Oxybutynin until toilet training versus observation after ablation	Improved hydronephrosis and VUR; no difference in serum creatinine, GFR, UTI incidence or successful toilet training; discontinued in 20% for side effects
Mirabegron (selective β_3 -adrenoceptor agonist)	Decreases detrusor overactivity and improves compliance	Early evidence of safety in children with PUV; further study needed
Botulinum toxin (bladder neck)	Hypertrophied bladder neck	Failed to improve voiding dynamics

- **What it is for:** the competing aetiological hypotheses point to **opposite drugs** — functional obstruction at the hypertrophied bladder neck argues for an α -blocker, bladder wall thickening from collagen deposition for an antimuscarinic.
- **Monitoring rule for oxybutynin:** if a growing child begins to show higher residual volumes and capacity, **stop it. Myogenic failure requiring CIC has occurred**, and it is unclear whether this reflects the natural evolution of the valve bladder or the drug.

Concept — the pharmacology of the valve bladder is undecided because the pathology is sequential: an α -blocker suits the early high-pressure outlet phase and an antimuscarinic the overactive phase, but both become harmful once hypocontractility supervenes, so the drug must follow the contractility pattern rather than the diagnosis.

Role of Urodynamics

- Bladder contractility is a dynamic condition changing with age, so serial monitoring rather than a single study is the point.
- Videourodynamics — fluoroscopy combined with standard urodynamics — is especially useful for assessing storage and emptying under dynamic conditions and for understanding the contribution of VUR to bladder function.
- Urodynamics is most useful when the clinical scenario changes — worsening upper tract dilatation or the appearance of UTIs.
- The counterweight is the known disadvantage of radiation exposure and infection.

Valve Bladder Syndrome

- Mitchell coined the term in 1982 after reviewing 11 patients in whom hydroureteronephrosis and renal function continued to worsen despite no clinical evidence of residual bladder outlet obstruction.
- The mechanism, as a sequence: initial compensation by high voiding pressures → rising urine volumes as the child grows, augmented by the polyuria of nephrogenic diabetes insipidus from evolving renal impairment → a bladder increasingly unable to empty completely → rising PVR → loss of any period of complete relaxation, with detrusor fibres continuously in partial or complete stretch → genotypic and phenotypic changes that further impair contractility → partial emptying immediately refilled by urine stored in the hydronephrotic kidneys → transmission of rising bladder pressure back to the kidneys → worsening renal function.

- **The three processes producing the valve bladder: polyuria, bladder insensitivity, and residual urine volume** — all secondary to **impaired bladder compliance creating high wall tension from increased collagen and matrix**, which together **sustain bladder overdistension**.
- **Valve bladder syndrome is a continuum of bladder dysfunction, not a discrete endpoint, and devolution into it is neither unavoidable nor uniform** — **effective drainage of the overdistended bladder may alter the course and preserve function**.

Treatment — the escalating ladder

- **Mainstay: monitoring of PVR, flow rates and voiding pressures, with timed voiding, double voiding, and anticholinergics or α -blockers.**
- **CIC becomes necessary if residual volume rises despite anticholinergics, to arrest progression toward decompensation.**
- **Overnight bladder drainage is an increasingly standard adjunct** in children with classic evidence of a developing valve bladder.
 - **Trigger: ureteric dilatation and hydronephrosis unresponsive to behavioural modification, or worsening renal function or recurrent UTI.**
 - **Method: a urinary catheter left in the bladder for a 7- to 10-hour period, giving extended bladder decompression so the kidneys drain without meeting the raised voiding or resting pressures they face by day.**
 - **It interrupts the vicious cycle of bladder remodelling and its renal consequences. Significant improvement in hydronephrosis, continence and UTIs after overnight drainage, intermittent nocturnal catheterisation and/or nocturnal double voiding.**
- **Where CIC or overnight drainage is difficult because of an elevated bladder neck or a sensate urethra, an appendicovesicostomy on the Mitrofanoff principle (1980) is the option.** After Mitrofanoff for a valve bladder, **pelvic ectasia decreased and compliance improved, yet renal deterioration still developed in 35%.**
- **Laparoscopic and robot-assisted creation of the catheterisable channel is increasingly adopted.**
- **Augmentation cystoplasty is now rarely used for a valve bladder, attributed to better understanding of bladder dysfunction, behavioural modification and timely overnight drainage.**
 - **Indication: a small-capacity, high-pressure, thick-walled valve bladder with worsening upper tract anatomy refractory to conservative measures.**
 - **Ureteral augmentation is preferred over ileal in PUV, avoiding the mucus production, acidosis and stones of ileal augmentation.**
 - **The ideal anatomy is already present in many of these boys — severe ureteric dilatation or unilateral VURD — allowing the ureter to be detubularised and patched onto a bisected bladder without bowel manipulation.**
 - **Lasting increases in capacity and reductions in filling pressure are reported at medium-term follow-up after ureteral augmentation.**

Concept — *every rung of valve bladder treatment buys the detrusor a period of genuine emptiness, from double voiding through nocturnal catheter drainage to a catheterisable channel, because the injurious variable is not pressure at a moment but the absence of any interval of relaxation.*

KEY POINTS

- Management of VUR in a child with valves **centres on the bladder**; ureteral reimplantation is rarely offered.

- The bladder evolves through three patterns: **detrusor hyperreflexia in infancy and early childhood; decreasing intravesical pressures with improved bladder compliance in childhood; increased capacity with hypocontractility and atony in adolescence.**
- **Videourodynamics is performed especially when there is clinical change** — increasing upper tract dilatation or recurrent UTI.
- **Overnight bladder drainage is considered for increasing PVR, UTI, or worsening hydronephrosis and renal function.**

PROGNOSTIC INDICATORS FOR RENAL FUNCTION

- Despite advances in antenatal diagnosis and intervention and in rapid postnatal treatment, the lifetime prevalence of end-stage renal disease (ESRD) in males with PUV is 20–50%.
- The five recognised risk factors: age at diagnosis, renal dysplasia with or without VUR, nadir creatinine during the first year of life, recurrent UTI, and bladder dysfunction.

Nadir Creatinine

- The 1-year value is more accurate as a predictive tool than the 1-month value.
- Nadir creatinine and creatinine velocity remain the best predictors of long-term renal function, with urinary peptides as a promising future avenue.
- Creatinine velocity — the rate of change of creatinine over the first 30 days of life — is itself a postulated predictor.

NADIR CREATININE THRESHOLDS COMPARED

Timing	Threshold	Implication
1 year	< 0.8 mg/dL	Minimal risk of ESRD
1 year	> 1.2 mg/dL	Higher risk of end-stage renal failure
1 month, post-treatment	< 0.8 mg/dL	Better long-term outcomes
1 month	> 0.85 mg/dL	Higher risk of renal failure
1 month	< 0.4 mg/dL	Lower risk

- The thresholds differ by time point and by whether the cohort was pre- or post-treatment; the 1-year value is the one to quote where both are available.

NADIR CREATININE BANDS AND ESTIMATED RISK OF END-STAGE RENAL FAILURE

Nadir creatinine (mg/dL)	Estimated risk of end-stage renal failure
< 0.4	0%
0.4 – 0.69	2%
0.7 – 0.99	27%
> 1.0	100%

- **What it is for:** it converts a continuous variable into a **step function**.
- The pattern is non-linearity — **0.7 mg/dL is the watershed**, where risk leaps from 2% to 27%.

- **The trap:** treating creatinine as a graded risk gradient and reassuring a family whose child sits at 0.7 rather than 0.69 — adjacent bands differ more than tenfold.

Concept — *renal prognostication in valves rests on a single number read at the right moment, and its power comes from being a threshold rather than a trend: the nadir creatinine reports what functioning nephron mass survived fetal development, which is why no biomarker has displaced it.*

Age at Presentation

- The assumption that antenatal diagnosis would forestall renal injury has not been sustained.
- Patients diagnosed in the pre-sonography era (before 1982) had a lower ESRD risk (16.8%) than those diagnosed in the post-sonography era (36.6%) — the paradox explained by **survival bias**: critically ill infants in the earlier era likely died before the diagnosis was complete, so modern survivors include a more severely affected cohort.
- **Poorer long-term renal outcome in those presenting before 1 year of age (41%) than after 1 year (15%)** follows the same pattern.
- **The countervailing assumption — that late presenters have milder disease — is also not corroborated:** some reports find **delayed presentation associated with significantly higher risk of azotaemia, higher serum creatinine and poorer long-term renal outcome.**

Concept — *the two directions of the age-at-presentation literature cannot both be read at face value because the denominator changed: earlier eras selected for survivors, and later series select for children whose disease declared itself late for reasons that may be severity rather than mildness.*

Quantifying Dysplasia Without Biopsy

- **Ultrasound features portending poor prognosis: hyperechogenic kidneys, cortical cystic change, and loss of corticomedullary differentiation.**
- **Renal parenchymal area — the area of the kidney minus the area of the pelvicalyceal system on the first postnatal ultrasound — is the quantitative development of this: each 1-cm² increase was associated with lower ESRD risk in infants with a 1-month creatinine in the intermediate band of 0.8 to 1.1 mg/dL.**
- **Parenchymal area differs significantly between children who progress to renal failure and those who do not (21.41 cm² vs 16 cm²), with a difference also in the corticomedullary differentiation index.**
- **Deep learning applied to ultrasound anatomic features predicted CKD progression, and was more accurate combined with existing clinical models than either alone;** machine learning may eventually allow office-based prognostic inference.
- **Other emerging predictors: fetal MRI, postnatal ultrasound findings, and genetic markers — all highlighting the need for tools incorporating genetic and biochemical markers.**
- **The VCUG is being revisited for prognosis: bladder height-to-width ratio and posterior urethra height-to-width ratio, with early evidence that a higher posterior urethral height-to-width ratio correlates with higher nadir creatinine.**

TRANSPLANTATION IN VALVE PATIENTS

- **ESRD prevalence in males with PUV is up to 50%; in the 2006 North American Pediatric Renal Trials and Collaborative Studies annual report, obstructive uropathy was the second commonest cause for transplantation, accounting for 15.8% of paediatric transplants since 1987.**
- **These recipients are a particularly difficult cohort:** likely comorbidities include **high-grade VUR into native non-functioning kidneys and valve bladder syndrome with a thick-walled, poorly contractile or hypercontractile bladder.**

- **The paediatric urologist must be a critical member of the transplant team** and examine the prospective recipient as part of pretransplant evaluation, because **complication and graft loss rates are significantly higher where bladder dysfunction is incompletely managed or the reservoir is not optimised.**

Outcomes

- **The thickened PUV bladder wall contributes to a significantly increased incidence of ureteric obstruction after transplantation** on both univariate and multivariate analysis, but **recent studies show no increased risk of graft loss or patient death despite ureteric obstruction, stenting or dilatation.**
- **Outcomes were similar whether initial management had been valve ablation, vesicostomy or supravescical diversion; bladder dysfunction increased graft failure risk but not significantly** (8-year follow-up).
- **Graft survival in PUV is similar to other CKD groups**, with pretransplant urodynamic assessment and bladder optimisation the requisite step.
- **Australian data: graft survival equivalent to other diagnoses until 10 years post-transplant, with increased graft loss thereafter** compared with other congenital anomalies of the kidney and urinary tract (CAKUT) — **attributed to bladder dysfunction.**

Pretransplant preparation

- **Videourodynamics should be obtained in every transplant candidate** to determine **safe storage pressures and contractile function of the future reservoir.**
- **Overnight bladder drainage or CIC may be started before transplant** to optimise the reservoir and establish the bladder management skills that graft success will require.
- **Pretransplant nephrectomy is rarely required** — considered only where **proteinuria or severe polyuria** creates haemodynamic challenges.

AUGMENTATION AND TRANSPLANTATION — THE SEQUENCING QUESTION

	Augmentation before transplant	Augmentation after transplant
Former dogma	Preferred , to avoid major reconstruction in an immunocompromised child, or one too young to manage catheterisation and pouch care	Avoided
Current evidence	Augmentation cystoplasty increases UTI risk in PUV, and where required, post-transplant outcomes are no different from pre-transplant	Acceptable; transplantation into a bladder managed by vesicostomy is a safe alternative until the child is old enough for cystoplasty and CIC
Infection	Augmentation did not compromise transplant kidneys, and infection rates were similar to non-augmented children provided catheterisation compliance was maintained	

- **What it is for:** it dismantles a dogma — the sequencing no longer carries an outcome penalty, so augmentation need not be forced before the child is ready.

Concept — *bladder function is the most important modifiable variable in the whole of valve transplantation: the graft itself performs normally, and every excess complication traces back to the reservoir it drains into, which is why urodynamic optimisation precedes the transplant rather than following it.*

KEY POINTS

- An antenatal vesicoamniotic shunt can be considered in selected patients with **bladder wall thickening, hydroureteronephrosis and oligohydramnios**; it may reduce the severity of pulmonary hypoplasia but **confers no protection from renal impairment.**

- A nadir serum creatinine at 1 year of less than 0.8 mg/dL confers a significantly decreased risk of developing ESRD.
- Transplantation into a valve-affected bladder may carry a **higher risk of ureteric obstruction, but no increased risk of graft loss** compared with controls.
- Graft survival is mediated by careful preoperative bladder evaluation and continuous long-term monitoring of bladder function.

QUALITY OF LIFE WITH POSTERIOR URETHRAL VALVES

- PUV is an illness with lifetime repercussions, and understanding long-term risk factors is necessary for counselling, preparing and treating these patients into adulthood.
- Developmental neurocognitive disorders appear more common in children with PUV: a Finnish registry-based study found neurodevelopmental disorder in 15% of males with PUV, with nearly 9% carrying an attention-deficit/hyperactivity disorder-attention-deficit disorder (ADHD-ADD) spectrum diagnosis — significantly increased against population controls.
- Early evaluation and early intervention once cognitive assessment is possible is part of overall management.
- Known associated risks of the valve bladder and of transplantation include erectile dysfunction and infertility.
- Lower urinary tract symptoms affect adults with a valve history two to three times more often than the general population.
- The apparent contradiction in the sexual-function data is one of subgroup: overall rates of erectile dysfunction and infertility in adult valve patients were no different from the general population, yet subgroup analysis of the same patients with urinary incontinence or renal insufficiency showed the greatest impairment of quality of life.
- The chronicity of bladder dysfunction, UTI risk and the sequelae of renal dysfunction require communication between paediatric and adult urologists, and the adult urologist must be conversant with PUV pathophysiology to care for these patients after transition.

Concept — the population-level reassurance and the individual-level burden are not in conflict: adults with valves do as well as anyone unless they carry incontinence or renal insufficiency, which is precisely why transition of care is targeted at that subgroup rather than at the diagnosis.

ANTERIOR URETHRAL VALVES

- The commonest congenital obstructive lesion of the anterior urethra, but 25 to 30 times less common than PUV.
- Frequently associated with a large anterior urethral diverticulum, the lesion being variously described as obstruction by a wall of the diverticulum or as a semilunar fold draping down from the anterior urethral wall.
- Embryology is unclear. Hypoplastic corpus spongiosum over the affected segment suggests either a place on the hypospadias spectrum or a faulty union between urethral mucosa and the epithelium of the fossa navicularis; rupture of dilated bulbourethral glands has also been proposed.
- Presentation varies with severity and hence with age: post-void dribbling and mild incontinence; significant bulging of the distal penis; a palpable obstructed bladder; or renal insufficiency and UTI.

Management, by scenario

- **Premature or small infant: vesicostomy** until the urethra will accommodate a cystoscope or further reconstruction.
- **Majority of cases: cystoscopy with valve ablation using a Bugbee electrode or laser, and when successful no further urethral surgery is required.**
- **Gross urethral diverticulum: surgical reconstruction over a urethral catheter.**

Outcomes

- As with PUV, long-term renal function is contingent on preoperative creatinine and GFR.
- Pretreatment azotaemia, VUR and UTI together increased the risk of poor renal outcome 25-fold.

ANTERIOR VERSUS POSTERIOR URETHRAL VALVES

	Posterior urethral valves	Anterior urethral valves
Frequency	Commonest cause of infant bladder outlet obstruction	25–30 times less common ; commonest <i>anterior</i> urethral obstructive lesion
Associated lesion	None analogous	Large anterior urethral diverticulum in many cases
Site	Posterior urethra, from the verumontanum	Bulbar urethra, penoscrotal junction or penile urethra
Clinical sign	Weak stream, palpable bladder, antenatal hydronephrosis	Bulging of the distal penis; urine expressed on compressing the distal shaft
Diagnosis	VCUG — dilated posterior urethra, abrupt funnelling	VCUG — dilated anterior urethra , bladder diverticula, massive VUR
First line	Endoscopic valve ablation	Endoscopic ablation, Bugbee electrode or laser — often definitive
Bladder dysfunction	Near-universal focus of lifelong care	Up to 80%
Renal outcome	ESRD 20–50%	78% normal renal function after treatment

- **What it is for:** the two share their endoscopic solution and their dependence on preoperative renal function, and differ only in how much nephron mass was lost in utero.

Concept — *the anterior lesion obstructs less completely and declares itself later, which is the single reason its renal prognosis is better despite an identical treatment principle.*

URETHRAL ATRESIA

- Also termed congenital urethral stricture; rarely described, probably because of its high associated mortality.
- **Survival depends on decompression having occurred:** the obstruction being **incomplete**, or decompression by an **antenatal shunt** or a **patent urachus**. **When an infant does survive, the outcome can resemble that of obstructing PUV.**
- **Anatomy:** an obstructing membrane at the distal end of the prostatic urethra, with the urethra distal to it often hypoplastic.
- **Part of the LUTO spectrum:** confirmed after birth by cystoscopy, and a vesicostomy is usually required.
- **The association with prune-belly syndrome remains controversial**, but infants often present with oligohydramnios, bilateral hydroureteronephrosis and weak abdominal wall musculature.

- **PADUA**, initially reported by Firlit, is considered a safe alternative for restoring urethral continuity in selected cases without complex reconstruction — long-term data are lacking.

Fig. 45.18. Cystoscopy on day 2 of life in a neonate with a prenatal history of lower urinary tract obstruction, demonstrating **near-complete urethral obstruction in the mid-bulbar urethra, most consistent with urethral atresia.**

Concept — *urethral atresia is selected for by decompression: every survivor has had a pressure escape route, which is why the reported phenotype resembles valves rather than the lethal lesion it is in utero.*

URETHRAL DUPLICATION

- A rare urethral anomaly with several anatomic variants.
- The duplication may begin at the bladder neck or more distally. One urethra usually terminates on the glans near its orthotopic position, while the other may open on the glans, more ventrally along the shaft, or, in the most severe cases, as proximally as the anal sphincter.
- The governing anatomical rule: duplication occurs in the sagittal plane, and the ventral urethra — usually the functional meatus — contains the sphincteric complex and the verumontanum.
- Diagnosis is obvious where **two distinct meatal openings are seen on the glans**; otherwise it requires a high index of suspicion in what appears to be atypical proximal hypospadias with a patent-appearing glanular opening.
- VCUg confirms the diagnosis during the voiding phase in many cases; retrograde injection of the distinct urethra is an alternative.

EFFMANN CLASSIFICATION OF URETHRAL DUPLICATION

Type	Anatomy
Type I	Blind, incomplete urethral duplication — an accessory urethra
Type II	Complete patent urethral duplication, with four subtypes
Type III	Urethral duplication as a component of partial or complete caudal duplication
"Y" type	Type IIA2 — the duplicated urethra arises from the first, then diverges to open at a second meatus as ventrally as the rectum

- **What it is for:** the type predicts the operation — type I is observed or coagulated, type II is reconstructed, and the "Y" variant requires staged transfer of the functional meatus.

Surgical management, by variant

- Small blind-ending accessory urethra → conservative monitoring, or simple coagulation of the mucosal tract with a Bugbee electrode.
- Patent duplicated urethra connected to the bladder with a distinct bladder neck → planned reconstruction.
- Two openings on the glans, the dorsal urethra mimicking distal epispadias → incise the septum between the two channels and repair the dorsal defect with reverse glans wings covering a dorsal urethroplasty.
- "Y"-type arrangement → staged reconstruction with preputial or buccal mucosa flaps, bringing the more ventral but functional meatus to the glans, where it is insinuated into the dorsal, usually atretic, urethra.
- Ventral functional meatus sited in the anal sphincter → anterior sagittal trans-ano-rectal approach (ASTRA) to mobilise it, the gap bridged by mobilising the ventral urethra and the remainder reconstructed with preputial skin.
- Where the distal aspect of the orthotopic (dorsal) urethra is of sufficient calibre, the ventral urethra may be mobilised enough for direct anastomosis to it.

- Treatment strategies vary significantly, and surgical strategy must be devised on individual anatomy.

Fig. 45.20. (A) Voiding cystourethrogram showing urethral duplication. (B) Intraoperative demonstration in the same child, with the larger-calibre ventral urethra and the narrower dorsal urethra.

Concept — *the reconstructive logic follows one anatomical fact: function lies ventrally with the sphincter and verumontanum while the cosmetically correct meatus lies dorsally, so every repair moves the functional channel into the orthotopic position rather than preserving the better-looking one.*

URETHRORRHAGIA

- Also called benign urethritis or idiopathic urethrorrhagia: spotting of blood on the underwear after urination, or voiding of clear urine followed by a few drops of blood.
- The history localises the lesion: blood in the underwear after voiding clear urine places the inflammation distal to the external sphincter.
- Typically benign and self-limited, though visibly alarming to families; commonest in peripubertal males.
- Proposed aetiologies: meatal stenosis and dysfunctional elimination syndrome; and, for the voiding-dysfunction school, a sequence — incomplete relaxation of the external urethral sphincter → increased voiding pressures → turbulent flow → negative intraluminal urethral pressure → engorgement of urethral mucosal sinuses → small extravasation of blood. Meatal stenosis is proposed to act by the same route.

Evaluation — *the paediatric departure from adult practice*

- Adult urologists regard cystoscopy and upper tract evaluation as necessary for haematuria; in symptomatic adolescent males, spontaneous resolution occurs in up to 92%, so routine cystoscopy is not necessary.
- The appropriate evaluation: a detailed history of bowel and bladder function, renal and bladder ultrasound, and office measurement of urinary flow rate and PVR.
- Cystoscopy is reserved for atypical urethrorrhagia — accompanied by symptoms of urethral stricture or by increased urethral bleeding.

The stricture association

- Although most urethrorrhagia is idiopathic, urethral stricture is diagnosed during or after evaluation in 14–28% of patients.
- This raises the question of whether cystoscopy itself, traversing abnormal inflamed epithelium, initiates stricture formation.
- In atypical cases, cystoscopy showed varying levels of bulbar urethral inflammation; about a quarter with a white membranous exudate were subsequently diagnosed with a stricture over a mean of 5 years, and 12% had a stricture found at cystoscopy with no prior instrumentation.
- The balanced reading: cystoscopy is benign in most cases, but may exacerbate inflammation in a small subgroup that no radiological modality can identify in advance — hence careful analysis of voiding habits and baseline uroflow before cystoscopy is considered.

Concept — *urethrorrhagia is managed as a voiding disorder rather than a haematuria, and the reason is the 92% spontaneous resolution rate set against a procedure that may itself propagate the stricture it is performed to exclude.*

URINARY FISTULA IN MALES WITH ANORECTAL MALFORMATION

- The vast majority of males born with an anorectal malformation have a fistula to the urinary tract.

- **The fistula is usually demonstrated on distal colostogram** during workup before anorectoplasty, the initial diverting colostomy having been performed in the neonatal period.
- **Associated genitourinary anomalies occur in 25–50%**, so routine diagnostic imaging for renal anomalies and VUR is recommended.
- **In a series of anorectal malformation patients, 16.3% developed a febrile UTI, of whom about half had VUR**; on multivariate analysis, the association of genitourinary malformation with UTI **did not reach statistical significance**.
- **Definitive repair is by a posterior sagittal approach, addressing the rectourinary fistula concurrently, with care to identify the tract and excise it in its entirety adjacent to urethra and rectum to prevent a diverticulum.**

Operative sequence, with the urologist present

- **Cystoscopy, attempting to pass an open-ended ureteric catheter through the fistula tract** — a guidewire may be passed if the fistula is of small calibre → **place a urethral catheter into the bladder across the urethra** → **position the child prone and begin posterior sagittal anorectoplasty** → **mobilise the rectum away from the urethra, using the previously placed urethral catheter as a guide to location** → **excise the fistula tract completely and approximate the urethral defect as close to the urethra as possible, reducing the risk of a urethral diverticulum** → **close the rectal defect primarily, or excise the fistulous portion of intestine where redundant tissue is available** → **interpose healthy adjacent tissue** → **leave the catheter for at least 1 week to ensure healing of the urethroplasty.**

Concept — *the catheter placed at the start of the anorectoplasty is the operation's safety device: it marks the urethra for the surgeon mobilising the rectum and then stents the repair, which is why the urologist's contribution comes before the dissection rather than after an injury.*

KEY POINTS

- **Anterior urethral valves and urethral atresia are less common causes of LUTO**, and long-term outcomes can be similar to those of PUV if the neonate survives.
- **Dysfunctional voiding causing turbulent urinary flow may lead to urethrorrhagia.**
- Treatment for urethrorrhagia should focus on **improving voiding habits and conservative management**, reserving cystoscopy for refractory cases and where there is significant straining during voiding.
- An infant diagnosed with anorectal malformation should undergo **renal ultrasound and VCUG** because of the high risk of associated genitourinary anomalies.

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